5–7 LPA Placement Preparation Sheet

1. Basic Coding Questions — 50 Problems

1. Reverse a String
2. Check if a String is a Palindrome
3. Check if a Number is a Palindrome
4. Reverse an Integer
5. Check if a Number is Prime
6. Print Prime Numbers in a Range
7. Find Factorial of a Number
8. Generate Fibonacci Series
9. Find the Nth Fibonacci Number
10. Check if a Number is Armstrong
11. Check if a Number is Perfect
12. Find GCD of Two Numbers
13. Find LCM of Two Numbers
14. Count Digits in a Number
15. Sum of Digits of a Number
16. Reverse Words in a String
17. Count Vowels and Consonants
18. Count Frequency of Characters
19. Find Duplicate Characters in a String
20. Remove Duplicate Characters from a String
21. Check if Two Strings are Anagrams
22. Find the First Non-Repeating Character
23. Find the Largest Element in an Array
24. Find the Smallest Element in an Array
25. Find the Second Largest Element in an Array
26. Find the Second Smallest Element in an Array
27. Find the Sum of Array Elements
28. Reverse an Array
29. Check if an Array is Sorted
30. Remove Duplicates from an Array
31. Find Duplicate Elements in an Array
32. Find Missing Number in an Array
33. Find Frequency of Array Elements
34. Find Common Elements in Two Arrays
35. Merge Two Sorted Arrays
36. Find Intersection of Two Arrays
37. Find Union of Two Arrays
38. Move All Zeros to the End
39. Find Maximum and Minimum Difference
40. Find Pair with Given Sum

41. Find Triplets with Given Sum
42. Implement Linear Search
43. Implement Binary Search
44. Implement Bubble Sort
45. Implement Selection Sort
46. Implement Insertion Sort
47. Find Maximum Subarray Sum
48. Rotate an Array by K Positions
49. Check if Parentheses are Balanced
50. Implement a Stack Using an Array

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2. Core CS Subjects to Prepare

Operating Systems (OS)

- Processes vs Threads
- Process Scheduling
- FCFS, SJF, Round Robin, Priority Scheduling
- Context Switching
- Deadlock and Deadlock Prevention
- Semaphores and Mutex
- Synchronization
- Paging and Segmentation
- Virtual Memory
- Page Replacement Algorithms
- Memory Management
- CPU Scheduling

Database Management Systems (DBMS)

- DBMS vs RDBMS
- Primary Key and Foreign Key
- Candidate Key, Super Key
- Normalization — 1NF, 2NF, 3NF
- SQL Queries
- Joins
- GROUP BY and HAVING
- Subqueries
- Aggregate Functions
- Indexing
- Transactions
- ACID Properties
- DELETE vs DROP vs TRUNCATE

- Views
- Stored Procedures
- Database Relationships

Object-Oriented Programming (OOPS)

- Class and Object
- Encapsulation
- Abstraction
- Inheritance
- Polymorphism
- Method Overloading
- Method Overriding
- Constructor
- Destructor
- Interface
- Abstract Class
- Access Modifiers
- Static Members
- Composition vs Inheritance
- Real-world OOP examples

Computer Networks (CN)

- OSI Model
- TCP/IP Model
- TCP vs UDP
- HTTP vs HTTPS
- IP Address
- IPv4 vs IPv6
- MAC Address
- DNS
- DHCP
- ARP
- Routing
- Switching
- LAN, MAN, WAN
- HTTP Methods
- Three-Way Handshake
- Firewalls
- Network Topologies

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3. HR Questions to Practice

1. Tell me about yourself.
2. Walk me through your resume.
3. Why do you want to join our company?
4. Why should we hire you?
5. What are your strengths?
6. What are your weaknesses?
7. Where do you see yourself in 5 years?
8. Why did you choose your engineering branch?
9. Tell me about your most important project.
10. What challenges did you face while building your project?
11. What was your specific contribution to the project?
12. Tell me about a time you solved a difficult problem.
13. Tell me about a failure and what you learned from it.
14. How do you handle pressure and deadlines?
15. How do you handle conflicts within a team?
16. Are you comfortable working in a team?
17. Are you comfortable relocating?
18. Are you comfortable working in shifts if required?
19. What motivates you?
20. Why should we select you over other candidates?
21. What are your short-term career goals?
22. What are your long-term career goals?
23. What do you know about our company?
24. What are your salary expectations?
25. Do you have any questions for us?

Preparation Priority

Coding → Core CS → Projects → Resume → HR

For a 5–7 LPA role, focus on solving basic-to-medium coding problems confidently, understanding the fundamentals of OS/DBMS/OOPS/CN, and being able to explain every project and technology mentioned on your resume.